Nizaal Khot

Senior Full-Stack & Agentic AI Systems Engineer

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Agentic AI & Orchestration | Full Stack Backend Architecture | MLOps & Production Engineering

PROFESSIONAL SUMMARY

Senior AI & Systems Engineer with 4+ years of designing and deploying scalable Agentic AI, RAG pipelines, and cloud-native backend architectures. Proven track record leading cross-functional AI initiatives, optimizing autonomous workflows with LangGraph, and deploying secure LLM microservices across AWS/Azure.

TECHNICAL SKILLS

- Core Technical & Management: AI/ML, Python, System Architecture, Software Development, Cloud Technologies, APIs, Database Design (SQL), Project Planning, Team Leadership, Code Review, Problem Solving.
- Agentic AI & Execution Frameworks: Multi-Agent Orchestration, Autonomous Planning & Execution Loops, Proactive memory & Tool-Use Systems, LangGraph, LangChain, DSPy, Advanced Prompt Optimization.
- ML, Forecasting & RAG: Contextual AI, Retrieval-Augmented Generation (RAG), Vector/Similarity Search (Vector DBs), Multi-Model Ensembles, CNN Image Classification.
- Infrastructure & MLOps: AWS (SageMaker, Bedrock, EC2, S3), Azure, GCP, Containerization (Docker, Kubernetes), CI/CD (Jenkins, GitLab), Real-Time Performance Monitoring, System Security & Reliability.

PROFESSIONAL EXPERIENCE

Dentsu Global Services | Mumbai, India

Senior Software Engineer (AI/ML Specialization) | Jul 2025 – Present

- Led AI/ML and technology projects from planning to implementation, defining system architecture, technical standards, and technical deliverables for production agent systems.
- Architected and deployed a hierarchical multi-agent ecosystem via LangGraph using 3 specialized autonomous agents (Execution, Review, and Feedback loops), reducing production cycle times by 86%.
- Developed and integrated AI/ML models, APIs, and automated document extraction solutions on AWS Bedrock, mitigating manual data entry workflows by 52%.
- Provided team leadership by guiding and mentoring AI engineers and developers. Established development strategies, conducted regular code reviews, and enforced rigorous coding and quality standards.
- Monitored system performance, security, scalability, and reliability, utilizing latency tuning and Root Cause Analysis (RCA) to eliminate pipeline bottlenecks.
- Collaborated closely with Management and business teams to convert business requirements into high-impact technical solutions and identify new AI automation opportunities.

Software Engineer (Generative AI & Data Pipelines) | Mar 2024 – Jul 2025

- Designed and scaled full-stack GenAI pipelines using GPT and Gemini models to build contextual software, automating production workflows and slashing operational overhead.
- Engineered robust, high-throughput ETL data pipelines to ingest, clean, and structure large-scale datasets across cloud warehouses.
- Evaluated and selected appropriate technologies, leading R&D on GitHub Copilot integration strategies adopted globally across the organization.

Associate Software Engineer & Consultant | May 2022 – Mar 2024

- Built containerized Python web scraping and data crawling pipelines optimized for cluster-based processing, streamlining massive data ingestion from e-commerce platforms.
- Integrated custom data pipelines seamlessly with enterprise APIs during rapid-prototyping Agile sprints.

PROJECTS HANDLED

- Production LLM RAG System Architecture: Designed and deployed a low-latency, context-aware RAG pipeline utilizing advanced vector similarity search and LangChain to deliver highly accurate contextual enterprise data retrieval.
- ML Model Microservice Deployment: Built, trained, and containerized a CNN model in TensorFlow, deploying it as a low-latency Dockerized REST API microservice for real-time validation pipelines.
- End-to-End Business Automation: Programmed full-stack business process automations via Power Automate and custom scripts, reducing manual operational effort across teams by 40%.

CERTIFICATIONS

- AWS Certified: Machine Learning – Specialty (Amazon Web Services)
- Microsoft Certified: Azure AI Fundamentals (Microsoft)

EDUCATION

Bachelor of Engineering in Information Technology (Graduated: 2022)
M.H. Saboo Siddik College of Engineering, Mumbai, India (CGPA : 8.48/10.00)